

KENTUCKY ROUTE 708 BRIDGE
Spanning Middle Fork of Kentucky River
Talleys vicinity
Lee County
Kentucky

HAER No. KY-15

HAER
KY-15

PHOTOGRAPHS

WRITTEN HISTORICAL AND DESCRIPTIVE DATA

HISTORIC AMERICAN ENGINEERING RECORD
SOUTHEAST REGIONAL OFFICE
National Park Service
U.S. Department of the Interior
100 Alabama St. NW
Atlanta, GA 30303

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KENTUCKY ROUTE 708 BRIDGE

HAER No. KY-15

Location: Spanning the Middle Fork of the
Kentucky River near Tallege, Lee
County, Kentucky

UTM: 17-269860-4159430
QUAD: Tallege

Date of Construction: 1917

Present Owner: Kentucky Transportation Cabinet
State Office Building
Frankfort, Kentucky

Present Use: Vehicular Bridge on KY 708

Significance: Only mixed pony-through truss by
Oregonia Bridge Company in Kentucky

Historian: Gregory D. Rawlings

KY 708 Tallege Bridge, Lee County

The KY 708 Tallege Bridge of the Middle Fork of the Kentucky River is eligible for the National Register of Historic Places as a structure of State and local importance. It possesses integrity of location, design, setting, materials, workmanship and embodies the distinctive characteristics of a type, period and method of construction. The 1917 metal truss bridge, although in poor structural condition, is a good example of the evolution of civil engineering technology and bridge construction in Kentucky. The current Structural Inventory and Appraisal (SIA) rating on the KY 296 Bridge is 16.6 out of a possible 100 points.

The bridge is located approximately 2000 feet south of the KY 52/KY 708 intersection and provides for traffic from several schools and single family residences along the valley floors. A Kentucky Geological Survey benchmark is located on the bridge indicating an elevation of 681 feet. The bridge is oriented on an east-west axis, with a perpendicular crossing of the river. The original bridge consisted of the main span Pratt Through truss flanked on each end by two Pratt Half-hip Pony trusses. The western end pony truss was replaced in 1983 by a bailey bridge after it failed.

The design of the main center span is that of a Pratt Through truss, the most popular late 19th and early 20th century bridge type. This type was used when the requirements for heavy loads and long

spans dictated the need for top lateral bracing with upper lateral struts and sway bracing for increased rigidity. The Pratt Through truss exhibits the greatest variation in refinements and embellishments. The remaining Pratt Half-hip Pony truss, the eastern span of the bridge, is without top lateral bracing. Pratt pony trusses were used for economical reasons when the weight and length requirements of particular crossing could be met by short spans of limited weight capacity. The combination of the bridge types makes this structure and interesting pin-connected bridge. This elegant truss rests high above the river on unique metal piers constructed of angles and lacing bars. This is the only mixed pony-through truss constructed in Kentucky by the Oregonia Bridge Company of Lebanon, Ohio.

The three spans of the bridge have an overall length of 255 feet and a deck width of 12 feet. The Pratt Through truss is 127 feet long, with a substructure composed of concrete abutments and metal frame truss piers. As with all bridges of this type, the top and bottom chords are parallel and the end post inclined. The end posts and top chords are built-up from 2 channels, cover plate and stay bars. The bottom chords are 2 rectilinear loop welded eyebars. The intermediate posts are two channels with 2 sets of lacing bars. The diagonals are 2 round eyebars and 2 round stirrup rods in the center panel and 2 rectilinear eyebars that are loop welded in the remaining panels. The hip verticals are built-up

from 2 angles and stay bars. One round eyebar that is loop-welded with a turnbuckle for field adjustments make up the counters. The top lateral bracing is a singular round rod and the top lateral struts are paired angles. The floor beams and stringers are rolled I-beams and the bottom lateral bracing consists of one round rod. Other details entail the use of curved portal bracing and original guardrailing.

The remaining Pratt Half-hip Pony truss is 66 feet long and also pin-connected. The end posts and top chords are 2 channels with a cover plate and stay bars underneath. The bottom chords are 2 rectilinear eyebars that are loop-welded. The intermediate posts are paired angles with lacing bars. The diagonals are 2 rectilinear eyebars that are loop-welded and the counters consist of one round eyebar that is loop-welded and filled with a turnbuckle. The floor beams and stringers are rolled I-beams and the bottom lateral bracings are singular round rods. Other details include a wood deck and original guardrail.